

Class 12 HW

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Q13. Read this file into R and determine the sample size for each genotype and their corresponding median expression levels for each of these genotypes. Sample size for each genotype: A/A - 108, A/G - 233, G/G - 121 Corresponding median expression levels: A/A - 31.25, A/G - 233, G/G - 121

How many samples do we have?

```
# Import data from file
expr <- read.table("rs8067378_ENSG00000172057.6.txt")
head(expr)
```

	sample	geno	exp
1	HG00367	A/G	28.96038
2	NA20768	A/G	20.24449
3	HG00361	A/A	31.32628
4	HG00135	A/A	34.11169
5	NA18870	G/G	18.25141
6	NA11993	A/A	32.89721

```
nrow(expr)
```

```
[1] 462
```

```
# Find the sample size of each genotype
table(expr$geno)
```

```
A/A A/G G/G
108 233 121
```

```
# Looking at the five-number summary for each genotype
summary(expr$exp[expr$geno == "A/A"])
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
11.40	27.02	31.25	31.82	35.92	51.52

```
summary(expr$exp[expr$geno == "A/G"])
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
7.075	20.626	25.065	25.397	30.552	48.034

```
summary(expr$exp[expr$geno == "G/G"])
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
6.675	16.903	20.074	20.594	24.457	33.956

```
# Find the median of each genotype
geno_boxplot <- boxplot(expr$exp ~ expr$geno, plot = FALSE)
geno_boxplot$stats[3, ]
```

```
[1] 31.24847 25.06486 20.07363
```

Lets make a boxplot

Q14. Generate a boxplot with a box per genotype, what could you infer from the relative expression value between A/A and G/G displayed in this plot? Does the SNP effect the expression of ORMDL3? The boxplot shows that those with the A/A genotype have higher median ORMDL3 expression compared to those with the G/G genotype, while those with A/G are in the middle. This suggests that the SNP is associated with ORMDL3 expression levels, where the A allele increases expression.

Lets make a boxplot

```
# Load ggplot
library(ggplot2)
```

```
# Create a box plot from the given data
ggplot(expr) + aes(geno, exp, fill=geno) +
  geom_boxplot(notch=TRUE)
```

